

Algebra & Calculus Practice

Synopsis: Math is an integral part of doing physics at any level. We handle equations or algebraic expressions. You are accustomed to doing algebra in your head, although perhaps you don't know it.

Imagine you are in a car driving on a highway at 60 mph. Your destination is 120 miles away. How many hours will it take to get there? You immediately say, 2 hours. To get this answer, you did an algebraic manipulation.

To see how, remember that speed is the distance covered divided by the time taken.

$$\text{Speed (miles/hour)} = \frac{\text{Distance (miles)}}{\text{Time (hour)}}, \text{ that is, } 60 \text{ (miles/hour)} = \frac{60 \text{ (miles)}}{1 \text{ (hour)}}$$

To calculate the time taken, you took the total distance of 120 miles and divided it by the speed, 60 miles per hour.

$$\text{Time (hours)} = \frac{\text{Distance (miles)}}{\text{Speed (miles/hour)}}, \text{ that is, } \text{Time (hours)} = \frac{120 \text{ (miles)}}{60 \text{ (miles/hour)}} = 2 \text{ hours}$$

Let us use symbols, Speed = s , Distance = d , and Time = t , and rewrite the above expressions:

$$s = \frac{d}{t} \quad \text{and} \quad t = \frac{d}{s}$$

I want you to notice what changed from the first to the second expression: s in the numerator ("upstairs") became the denominator ("downstairs"), and t in the denominator ("downstairs") became the numerator ("upstairs") when s and t "crossed" over to the other side of the equal sign.

$$\frac{s}{1} = \frac{d}{t} \longrightarrow \frac{t}{1} = \frac{d}{s}$$

Note that any number divided by 1 is itself, hence $s = s/1$.

So, this particular manipulation involved two simultaneous "crossing overs." But it doesn't have to be. Let's just cross the t over and leave s .

$$\frac{s}{1} = \frac{d}{t} \longrightarrow \frac{st}{1} = \frac{d}{1}$$

Or $d = st$. In other words, you just derived and expressed the distance covered, d , when you travel at speed, s , for time, t . Using the example above, $(60 \text{ miles/hour}) \times (2 \text{ hours}) = 120 \text{ miles}$.

This "trick" — when you cross over the equal sign, you go from downstairs to upstairs and vice versa — will always work because it shortcuts the many steps of multiplying or dividing both sides that you are used to doing. You should convince yourself that you would get the same answer if you just did it the way you were taught in middle school. Don't bother with middle school procedures once you are convinced and comfortable with this method.

There is a similar shortcut involving addition and subtraction. Again, you are used to doing the algebra in your head. Imagine you are in a class of 25. There are 10 males. How many females are there? You immediately say 15. What algebraic manipulation did you do?

Let total = t , males = m , and females = f , so $t = m + f$. That is $25 = 10 + f$. To find f , you did $25 - 10 = 15$, the female number. Or algebraically, $f = t - m$.

The similar trick I am referring to is this: If you take a term (t , f , and m are the terms) and "throw it over" to the other side of the equal sign, it flips the sign.

$$(+t) = (+m) + (+f)$$

$$(+t) - m = (+f)$$

Note that a number without an explicit negative sign in front is a positive number. Let's do something weird. Let's throw the t over to the other side:

$$(+t) = (+m) + (+f)$$

$$0 = (+m) + f - t$$

Note that you must write a zero on the left-hand side when there are no terms or numbers! What does this look like with numbers?

$$0 = (+m) + f - t$$

$$0 = 10 + 15 - 25$$

You see that $0 = m + f - t$ makes total sense. The key message is this: If you throw a term over, positive becomes negative and vice versa.

Let's try something that has both addition/subtraction and multiplication/division with multiple terms.

$$d = d_0 + vt$$

Let's solve for v (or isolate v or make v the subject of the equation). There are two terms (d_0 and vt) on the right-hand side and one term (d) on the left-hand side (remember the terms are separated by + or - signs). If the symbols are multiplied or divided, they are grouped as being in a single term. Hence, vt is a single term consisting of v and t being multiplied.

Here is the sequence of operations you ought to follow:

1) **Isolate the TERM that contains the symbol of interest.** That is, "throw over" all terms you are not interested in solving. Here, you need to isolate the vt term, so throw d_0 over and switch the sign of d_0 .

$$d = d_0 + vt$$

$$-d_0 + d = +vt$$

2) **Divide out the FACTOR you do not want.** Here that factor is t . BUT BEFORE YOU DIVIDE OUT, BRACKET TOGETHER ALL TERMS ON THE OPPOSITE SIDE.

$$\frac{(-d_0 + d)}{1} = \frac{+vt}{1} \longrightarrow \frac{(-d_0 + d)}{t} = \frac{+v}{1} \longrightarrow v = \frac{d - d_0}{t}$$

The last line is the same expression written more neatly.

Practice: Algebra

Solve the following equations.

1) Solve for a. $c = b/a$	5) Solve for x. $z = 2x + y$
2) Solve for M. $D = M/V$	6) Solve for y. $z = 2/y + x$
3) Solve for f. $T = 1/f$	7) Solve for z. $x = y - 1/(2z)$
4) Solve for T. $\omega = (2\pi)/T$	8) Solve for t. $2x = (1 - 2y)/(2 - 3t) - 1$

One New Idea: The Derivative as a Rate of Change

PHY 151 is a *calculus-based* course, and here is the entire reason why: physics is the study of **change**, and the derivative is the mathematical machine that measures change. On the road trip, your speed was the rate at which your position changed — $s = d/t$ works when the speed is steady. When the speed is *not* steady, we shrink the time interval until it is instantaneous, and that limit is the derivative:

$$v = \frac{dx}{dt} \quad (\text{"velocity is the derivative of position with respect to time"})$$

For nearly everything this semester, one rule does the differentiating — the **power rule**, applied **term by term**:

$$\frac{d}{dt} (t^n) = nt^{n-1}$$

In words: the exponent hops down front as a multiplier, and the exponent drops by one. Two consequences you will use constantly: the derivative of a **constant** is zero (a constant doesn't change, so its rate of change is nothing), and a constant **coefficient** just rides along in front.

Let's differentiate the position equation from the Algebra Review Sheet, term by term:

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2$$

1) **Constant** → 0. x_0 is a fixed number; it does not change with time, so $\frac{d}{dt} (x_0) = 0$.

2) **Power rule**. $t = t^1$: the 1 hops down, the exponent drops to 0, and $t^0 = 1$, leaving just the coefficient:
 $\frac{d}{dt} (v_0 t) = v_0$.

3) **Power rule again**. The 2 hops down and cancels the $\frac{1}{2}$; t^2 becomes t^1 : $\frac{d}{dt} (\frac{1}{2}at^2) = at$.

4) **Collect the pieces**:

$$v(t) = \frac{dx}{dt} = v_0 + at$$

Differentiating position produced the *velocity* equation — equation 4 on the Algebra Review Sheet! Differentiate once more and only at survives:

$$a = \frac{dv}{dt}$$

Position → velocity → acceleration, one derivative at a time.

Going the *other* way — from acceleration back to velocity, or velocity back to position — is **integration**, which undoes the power rule (the exponent goes *up* by one, and you divide by the new exponent). We will build that up carefully in Module 1; for now, being comfortable with the derivative direction is all you need.

Practice: Calculus

Every one of these is the power rule, term by term.

9) Differentiate. Find $v(t) = dx/dt$. $x(t) = 5t^3 - 2t$	11) For the position below, find the time at which the velocity is zero. $x(t) = t^2 - 6t + 4$
10) Differentiate. Find $a(t) = dv/dt$. $v(t) = 4t^2 + 7$	12) Differentiate — from memory this time, without peeking above. $x(t) = x_0 + v_0 t + \frac{1}{2}at^2$